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Direction of Magnetization in Icelandic Basalts

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In the previous paper T. Einarsson has given the results of a magneto-geological study, carried out with a field compass, which showed that groups of normal and reversed magnetic polarity, with respect to the present geomagnetic field, alternate in a regular geological sequence in the Tertiary and Quaternary basalts of Iceland. Following this field investigation, oriented samples have been taken for laboratory measurements.

The magnetization is measured on an induction magnetometer built by A. Brynjolfsson (described in following paper), in which a sample of 1–2 kg rotates at a speed of 5 revolutions per second. The direction of magnetization is determined with an accuracy of 1°. Measurements on magnetization are carried out both before and after partial demagnetization in an alternating field of 110 oersted amplitude.

For statistical treatment of measured directions of magnetization the method of R. Fisher (1953) has been used. The frequency density on the unit sphere is accordingly assumed proportional to $\exp(k \cos\theta)$, where θ is the angular distance from the true direction and k a measure of precision. The mean direction for N observations is the direction of the sum, R, of all unit direction vectors and the probable error, $\bar{p}$, for this direction is found from the equation

$$1-\cos\bar{p}=\frac{N-R}{R}(2^{1/(N-1)}-1).$$

The precision index, k, is calculated as

$$k=\frac{N-1}{N-R}.$$

To find the probable angular error of single observations, p, we use the equation

$$1-\cos p=\frac{\ln 2}{k}=\frac{N-R}{N-1}\ln 2.$$

Most of the samples are taken in the Hvalfjordur district, near Reykjavik, where the field observations have been carried out in greatest detail. The present material only covers the six latest magnetic groups, three of normal and three of reversed polarity. The samples are, as far as possible, taken from open profiles, where the geological structure is clearly visible. The sites for collecting samples have been chosen on basis

of the magneto-geological field observations, and samples taken from within each magnetic group, as well as at the boundaries between groups of opposite polarity. Special attention has been given to places, where the field observations gave an indication of a gradual transition at the boundaries. In most cases only one sample was taken from each basaltic layer or lava bed. If more than one sample was taken the average value of magnetization is used here and treated as a single observation.

Samples from 33 lava beds belonging to the three groups of normal polarity show a distribution in directions of magnetization which corresponds to $k=38$ or a probable angular error for single samples amounting to 11°. Demagnetization reduces the probable error to 9°. Samples from 26 lava beds belonging to the three groups of reversed polarity show a distribution with a probable error of 24° before and 11° after demagnetization. The big scattering for the reverse magnetized samples in the natural state is probably partly due to isothermal remanent magnetization acquired in the earth's magnetic field. This secondary magnetization can even be so strong that certain samples from a reverse polarized rock formation show normal direction of magnetization in the natural state, but during demagnetization this turns to the regular reversed direction. A general feature for reverse magnetized samples is that the intensity of magnetization increases in the first stages of demagnetization. All this is easily explained by assuming an isothermal remanent magnetization which is less stable regarding demagnetization than the original thermoremanent magnetization. We therefore believe that in the demagnetized state the samples are in better agreement with the original magnetization than in the natural state, and from now on we only report results of measurements on demagnetized samples.

The measured direction of magnetization is corrected for geological dip, which in most cases is 4° or less. This direction is then interpreted as the direction of the geomagnetic field at the time when the rock was formed. Assuming that this field was a dipole field, we then calculate the geographic position of the magnetic poles, using the relation

$$\operatorname{tg} \phi = 2 \cot I$$

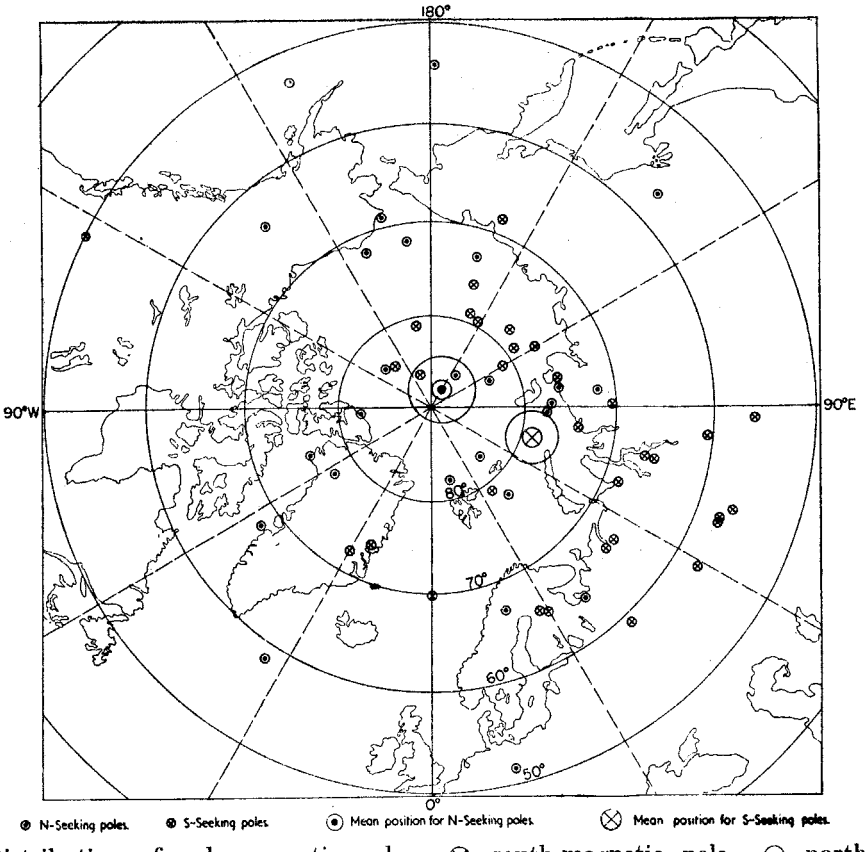
where ϕ is the angular distance to the south magnetic pole and I the magnetic inclination, positive for downward field directions and negative for upward directions. Instead of declination and inclination we then use the geographic position of one of the magnetic poles to represent the direction of magnetization in the samples. This seems to be a reasonable representation as it facilitates the comparison of results obtained at different parts of the globe.

In fig. 1 we see the results of measurements on 33 samples of normal and 26 samples of reversed polarity. For normal samples the positions of south magnetic poles are indicated by crosses, but for reversed samples positions of north magnetic poles are shown as dots. The magnetic poles are distributed around the geographic pole down to a latitude of 50°.

For south magnetic poles the distribution corresponds to $k=19$ or a probable error of $p=17°$. The corresponding values for north magnetic poles are $k=15$ and $p=18°$.

To compare this picture with the present geomagnetic field, the dipole formula has been used to calculate positions of magnetic poles from geomagnetic data of 20 magnetic observatories scattered over the earth.

Fig. 1



Distribution of paleomagnetic poles. ⊗ south-magnetic pole. ○ north-magnetic pole. The big marks show average positions with their probable errors drawn as circles.

The scattering of these poles due to present anomalies in the earth's field corresponds to a probable error of 8°. In the course of time the magnetic field is believed to change in such a way that the geographic pole is the average position for the magnetic poles. To get a measure for the expected scattering of magnetic poles over long time intervals it is therefore reasonable to choose the geographic pole as a centre for the distribution. This results in a probable error of 12°. As the scattering in the paleomagnetic data is 50% greater than this, it might either mean

that the present deviation of the magnetic pole from the geographic pole is less than average, or that our results show a greater scattering than corresponds to the state of the geomagnetic field in the past. If this is the case the scattering might perhaps be reduced by improved sampling, measuring or selection technique. Here no selection has been made and all measured samples are included in the results. By leaving out samples that have suffered big changes in magnetization during demagnetization, some reduction in the scattering of reversely magnetized samples can be obtained, but at the same time the number of valid samples is reduced so much that the accuracy of the average direction is lowered.

The mean position for all south magnetic poles is 73·9° E, 77·3° N with a probable angular error of 2·8°. We thus find a statistically significant deviation of the average south magnetic pole from the geographic pole. The average position for all north magnetic poles is 149° E, 87·8° N, with a probable error of 3·6°, in good agreement with the geographic pole as a centre for the distribution. Such agreement has also been found by Hospers (1954) in Icelandic Tertiary basalts of both polarities.

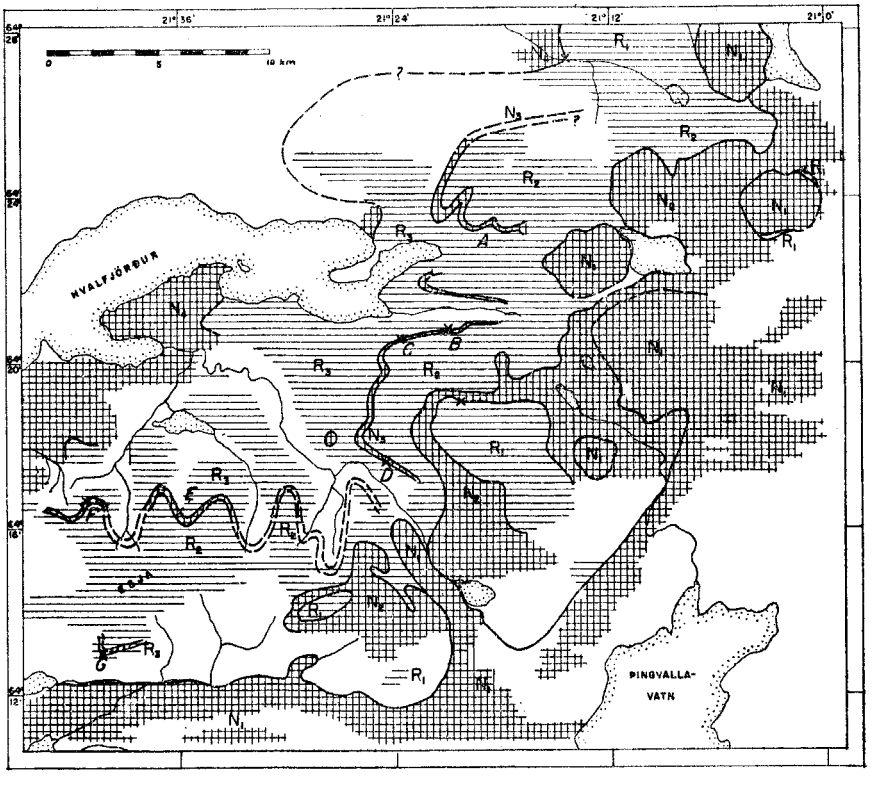
In order to investigate the origin of the deviation the results have been divided into the 6 magnetic groups. Only one of these groups, the oldest normal group, N_3, shows a significant deviation from the geographic pole. This group, however, is rather thin and only contains a small number of basaltic layers. It is therefore possible that the actual number of independent samples is not as great as assumed, and furthermore it is likely that some of these samples are from the very beginning of this normal period when the directions of the magnetic field may have been somewhat abnormal. For these reasons the observed deviation of the average magnetic pole from the geographic pole may not be the result of a polar wandering. This will first become clear when older samples are measured.

Figure 2 is a magneto-geological map of the area where most of the samples were taken. It is a part of T. Einarsson's map showing the Hvalfjordur district. The thin N_3 group stretches over most of this area. In seven places, marked with the letters *A* to *G*, samples have been collected. In all these places intermediate directions of magnetization were found at the lower boundary of the N_3 group. These intermediate directions make large angles with the present direction of the magnetic field and correspond to a magnetic pole in the neighborhood of the equator. In four more places intermediate directions have been found at the N_4–R_3, R_2–N_2 and N_2–R_1 boundaries. Three of these places are shown as crosses on the map, but one of them (R_2–N_2) is outside this area. We here use the term ' intermediate direction ' only for directions of magnetization which correspond to a magnetic pole with a latitude lower than 50°. Such intermediate directions have not been found within a magnetic group, but only at the boundary between two groups of opposite polarity. Even at such boundaries it is rather exceptional to

find intermediate directions. In most cases the magnetic groups are sharply divided, with the abrupt change in polarity occurring across a thin bed of sediments.

Altogether intermediate directions have been found in 45 lava beds. This number, however, is no measure for the frequency of occurrence of these directions in the basaltic rock formation, as samples were preferably

Fig. 2



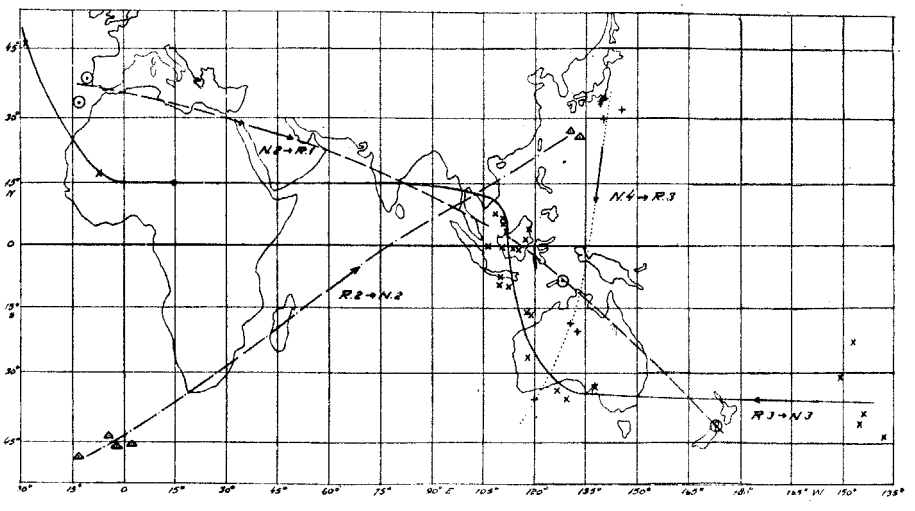
A magneto-geological map of the Hvalfjordur district.

taken at places where the field studies had given some indication of an intermediate direction. A rough estimate indicates that rocks having intermediate directions of magnetization may amount to a few per cent of the total rock formation.

The results of measurements on samples showing intermediate directions of magnetization have been treated in the same way as for the other samples. The direction of magnetization is corrected for geological dip and geographic position found for the south-magnetic pole of the corresponding dipole field. Figure 3 shows the calculated positions of south-magnetic poles corresponding to all the 45 intermediate directions. The results from each of the four transitions are distributed

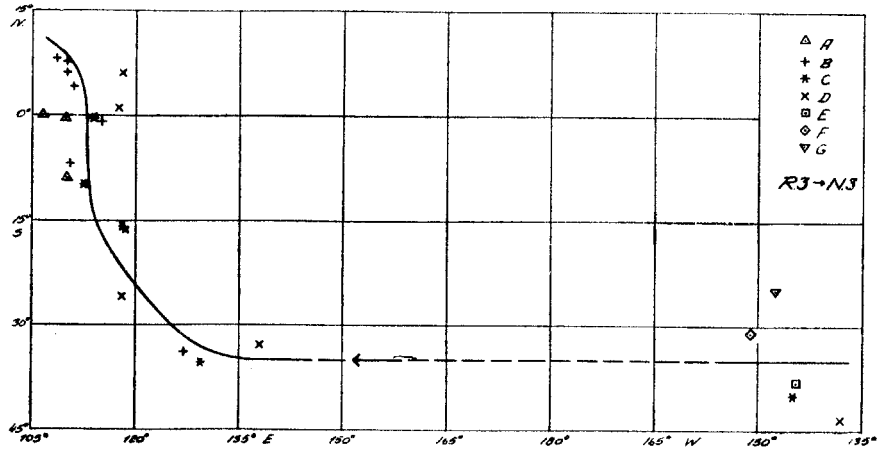
rather regularly, indicating a definite path followed by the magnetic pole during the reversal of the magnetic field.

Fig. 3



South-magnetic poles corresponding to intermediate directions of magnetization found at four different boundaries between magnetic groups.

Fig. 4



South-magnetic poles corresponding to intermediate directions of the R_3–N_3 transition.

Only the R_3–N_3 transition has been studied in any detail. Figure 4 gives the results of 24 determinations from this transition on samples taken at the seven locations marked from *A* to *G* on the map. These places are spread over a distance of 25 km. All these results fall on a

smooth curve to within 10° and the sequence of points along the curve agrees with the position of the samples in each profile. It seems reasonable to assume that this curve represents the transition of the geomagnetic field from the reversed to the normal direction. It is difficult to make an estimate of the time taken by this transition, but from the appearance of the basaltic layers, without any sediments between them, we should not expect that the time could be more than a few thousand years or even shorter.

Our results show some indication of a reduced field strength during reversals of the magnetic field. The mean magnetization of all normally and reversely magnetized samples is about 2×10^{-3} e.m.u./g, while for samples showing intermediate directions it is only half as much. Individual variations are, however, quite large so this can, in the best case, only give a very rough idea about the strength of the magnetic field. According to W. Elsasser (1956) the dipole term of the geomagnetic field should disappear during reversals. All we can say is that our results do not disagree with this statement.

If the dipole field disappears and only higher order harmonics are left during reversals we can not expect world wide agreement for the route of the calculated magnetic pole, but the intermediate directions may still prove quite useful for establishing geological correlations over large distances. The fact that we have found intermediate directions at 4 boundaries out of 6 in our investigation gives reason to believe that intermediate directions of magnetization will also be found in other basaltic regions. Correlation with sediments might also be a possibility. Some difficulty for this type of geo-chronological correlation might be caused by the apparent tendency of the calculated pole to pass the equator in a certain region.

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